

Invo Match

Automated accounts-payable invoice extraction, matching & validation pipeline

n8n

Node.js / Express

Google Gemini

NocoDB

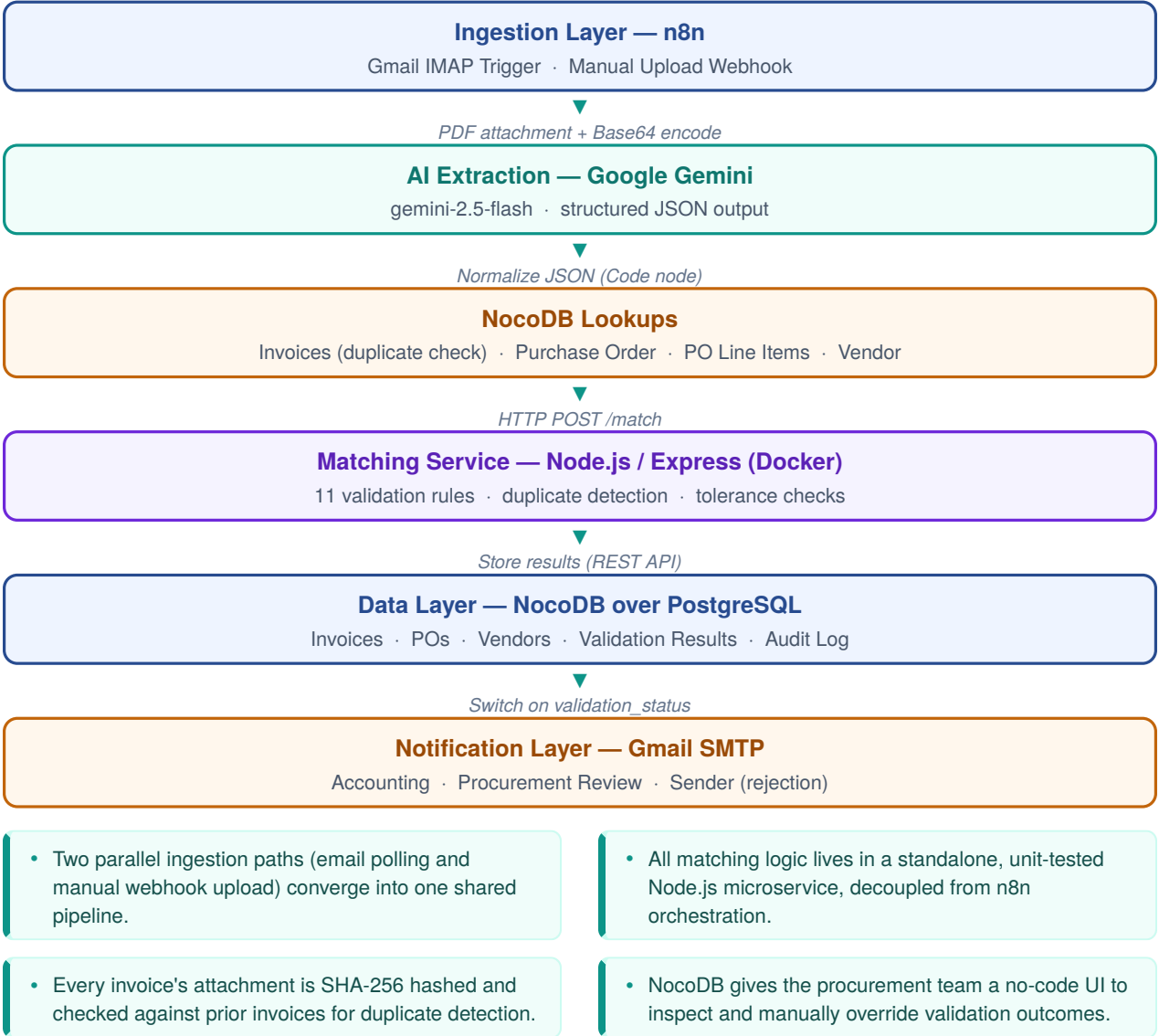
PostgreSQL

Docker

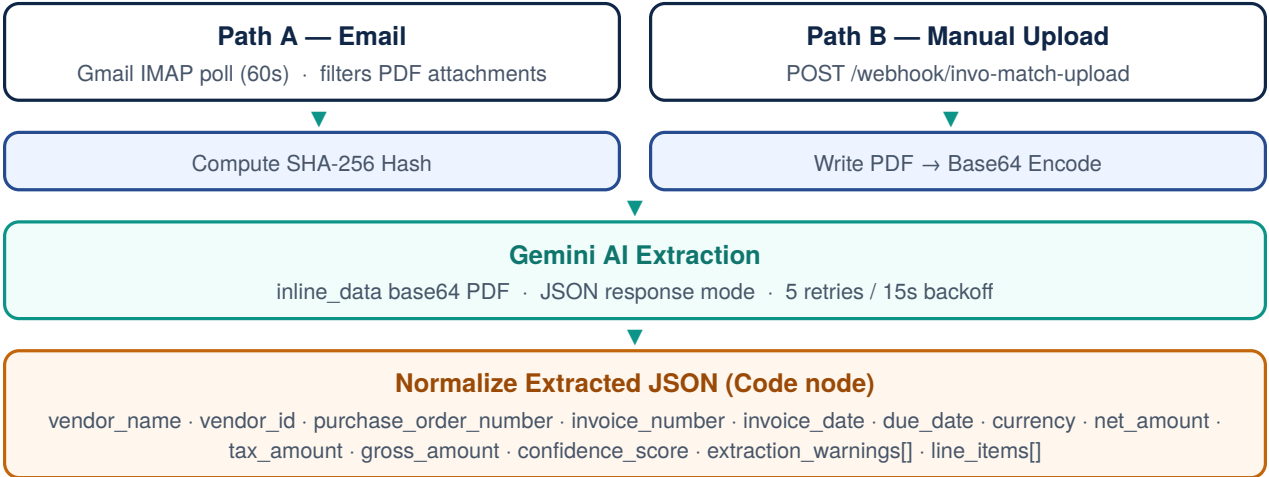
1 System Overview

Invo Match automates accounts-payable invoice processing: an incoming invoice PDF (by email or manual upload) is extracted with AI, matched against its Purchase Order and vendor master data, validated against eleven business rules, and routed to the correct outcome — ready for payment, procurement review, or rejected. The system is fully self-hosted, orchestrated by n8n, with a Node.js matching microservice and a NocoDB / PostgreSQL data layer, and requires no paid SaaS beyond the Gemini API.

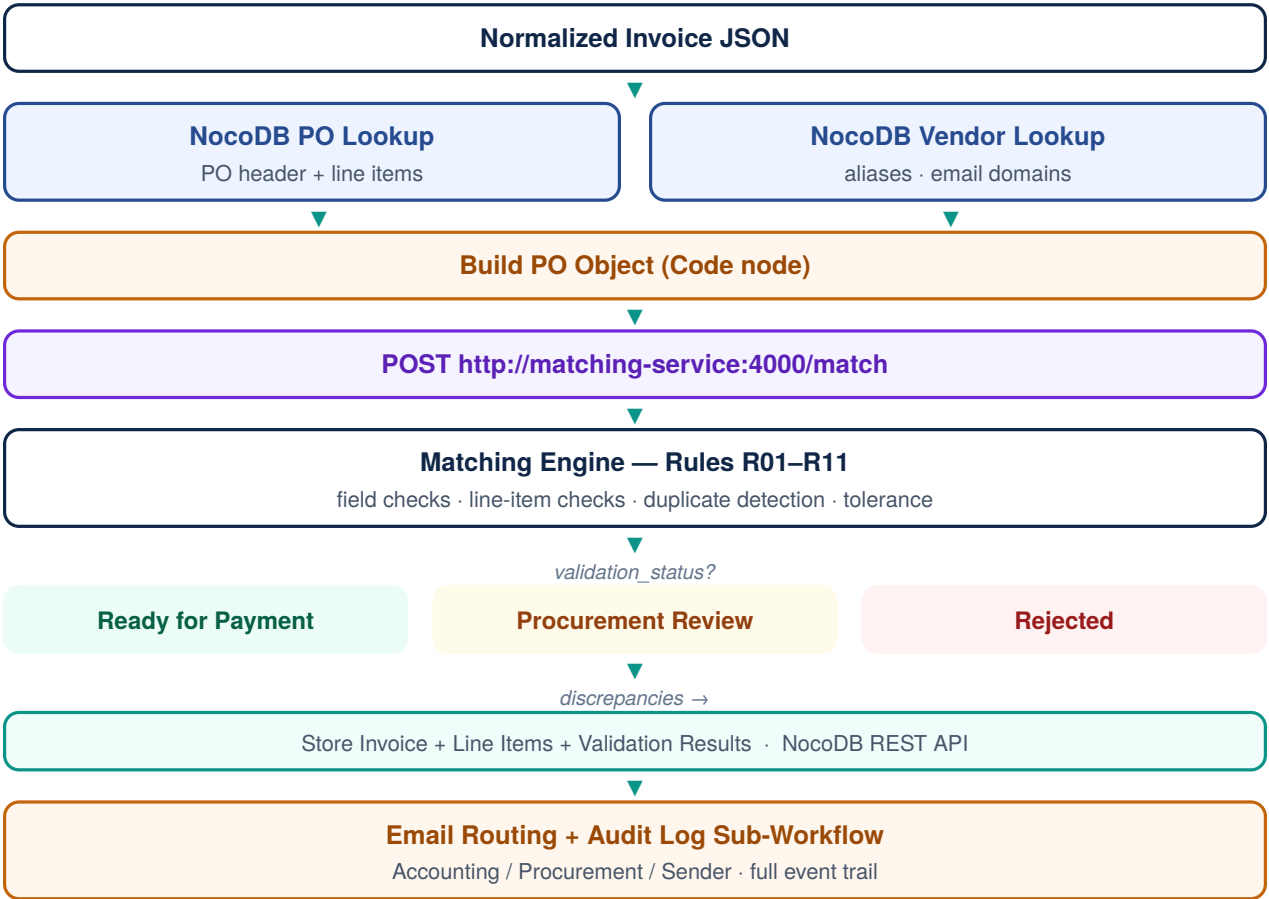
2 High-Level Architecture



3 Shared Ingestion & Extraction Pipeline



4 Matching & Routing Flow



5 Component Map

The codebase separates orchestration (n8n workflows), business logic (Node.js matching engine), and storage (NocoDB/PostgreSQL) so each layer can be tested and redeployed independently.

Module / Path	Primary Role	Key Details
workflows/invo-match-main.n8n.json	Email Ingestion Workflow	Polls Gmail IMAP every 60s, filters PDF attachments, feeds the shared pipeline.

Module / Path	Primary Role	Key Details
workflows/invo-match-manual-upload.n8n.json	Manual Upload Workflow	Webhook endpoint that reads a PDF from a local Docker volume and joins the shared pipeline.
workflows/procurement-review-webhook.n8n.json	Review Override Webhook	Lets procurement approve/reject a flagged invoice and writes the decision back to NocoDB.
workflows/log-audit-event.n8n.json	Audit Sub-Workflow	Called at each pipeline stage to append an entry to the audit_log table.
workflows/error-handler.n8n.json	Error Handler	Global n8n error trigger; captures failures and notifies via email.
src/matching.js	Matching Engine	Core validation logic: rules R01–R11, tolerance checks, string similarity, duplicate detection.
src/normalize.js	Normalization Helpers	Vendor name normalization, currency normalization, amount parsing.
src/schema.js	Shared Enums	ValidationStatus, Severity, RuleCategory, DuplicateType constants used across the engine.
src/server.js	Matching API	Express server exposing POST /match; wraps the engine for n8n's HTTP node.
src/setup_nocodb.js	DB Bootstrap Script	Provisions NocoDB tables/fields programmatically from the schema definition.
schema/database-schema.sql	PostgreSQL DDL	Table definitions for vendors, POs, line items, invoices, validation results, audit log.
prompts/invoice-extraction-prompt.md	Gemini Prompt	Zero-shot extraction schema/prompt sent with each invoice PDF.

6 Technology Choices

Layer	Choice	Why
Orchestration	n8n (self-hosted)	Visual workflow engine; native IMAP/webhook/HTTP/email nodes; free self-hosted tier.
AI extraction	Google Gemini (gemini-2.5-flash)	Accepts PDF as base64 inline data directly — no separate PDF parsing library needed.
Matching runtime	Node.js + Express	Lightweight, testable microservice; keeps business rules out of n8n Code nodes.
Database / UI	NocoDB over PostgreSQL	No-code UI for procurement team; full SQL underneath for reliable queries.
Storage	Local Docker volumes	No invoice PDFs leave the host; simplest compliant option for AP documents.
Deduplication	SHA-256 attachment hash	Cheap, deterministic duplicate detection independent of invoice number formatting.
Notifications	Gmail SMTP	Reuses the same Gmail account already used for IMAP ingestion; zero extra setup.
Testing	Node built-in test runner	node --test over normalize/matching unit tests plus an integration test suite.
Containerization	Docker + docker-compose	Reproducible multi-service stack: n8n, two Postgres instances, NocoDB, matching service.
PDF fixtures	pdfkit	Generates the sample invoice PDFs used for local demos and tests.

7 Validation Rules (R01–R11)

Rule	Check	Severity	Outcome Trigger
R01	PO exists	Critical	Rejected if missing
R02	PO approved	Critical	Rejected if not approved
R03	Duplicate detection (hash + invoice number)	Critical	Rejected if duplicate
R04	Vendor match (with alias support)	Major	Rejected if mismatch
R05	Currency match	Major	Rejected if mismatch
R06	Total within PO remaining budget	Critical	Rejected if exceeded
R07	Net / gross amount match	Major	Rejected if mismatch
R08	Tax calculation correctness	Minor	Review if wrong
R09	Line item count match	Major	Rejected if extra/missing
R10	Line item price match (per-item, with tolerance)	Minor	Review if within tolerance
R11	Confidence score threshold (< 0.75)	Minor	Review if below threshold

8 Scalability Considerations

Concern	Current Approach	Scale-Up Path
Invoice volume	Sequential n8n polling	n8n queue mode with multiple workers
AI rate limits	5 retries, 15s backoff	Gemini batch API or multiple API keys
Database	Single PostgreSQL instance	Read replicas, connection pooling
Matching service	Single Docker container	Horizontal scaling behind a load balancer

9 Security & Evaluation

Security Measures

✓ Credential handling API keys stored in .env, injected as Docker environment variables.
✓ Authenticated APIs n8n API secured via JWT token; NocoDB REST calls use API token auth.
✓ Local-only storage No invoice PDFs stored in the cloud — all data stays on local Docker volumes.
✓ Integrity check SHA-256 attachment hashing for duplicate and tamper detection.

Known Limitations

! Single matching-service instance No built-in horizontal scaling yet; add a load balancer for higher volume.
! Gemini dependency Extraction quality depends on a single external AI provider with no local fallback model.
! Sequential email polling 60s IMAP poll interval adds latency versus a push-based ingestion path.
! Manual override trail Procurement overrides are logged but not yet re-validated against updated PO data.